North East University Bangladesh

Department of CSE

Course Code: CSE-460

Course Title: Deep Learning Lab

Project Title: Ingredient Analysis System for Personalized Dietary Restrictions

Submitted To

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Respected Sir,

I am submitting my project proposal titled **“Ingredient Analysis System for Personalized Dietary Restrictions”** as part of the course requirements for Deep Learning. The project focuses on identifying potential allergens and intolerance-triggering ingredients from food labels using Optical Character Recognition (OCR) and Natural Language Processing (NLP) techniques.

Introduction:

Food allergies, intolerances, and dietary restrictions affect millions of people worldwide. Many individuals experience health problems after consuming packaged foods because ingredient labels are difficult to understand or contain hidden allergens and technical ingredient names unfamiliar to normal consumers.

For example:

- A person with lactose intolerance may unknowingly consume products containing whey or casein.
- A gluten-intolerant person may react to semolina or modified wheat starch.
- People with diabetes may consume products containing hidden sugars such as fructose syrup or maltodextrin.
- Individuals with nickel or salicylate sensitivity may face difficulty identifying risky ingredients from food packaging.

Understanding food labels often requires nutritional or biochemical knowledge, making it difficult for users to quickly determine whether a packaged product is safe for them.

This project proposes an AI-powered web application that uses Optical Character Recognition (OCR), Natural Language Processing (NLP), and Machine Learning (ML) to analyze food ingredient labels and determine whether a food product may be harmful for users with specific allergies, intolerances, or dietary restrictions.

Problem Statement:

Many packaged food products contain:

- complex ingredient names,
- hidden allergens,
- ambiguous food additives,
- multilingual labels,
- and difficult-to-read nutritional information.

Consumers with dietary restrictions often rely on manually reading labels, which can:

- be time-consuming,
- lead to mistakes,
- and potentially cause serious health issues.

Currently, there are limited intelligent systems capable of:

- automatically extracting ingredient text from food packaging,
- analyzing ingredient composition,
- and providing personalized health risk assessments.

Therefore, there is a need for an automated intelligent system that can help users identify whether a food product is safe according to their medical or dietary restrictions.

Aim of the Project:

The main aim of this project is to develop an intelligent food analysis system capable of:

- scanning packaged food labels,
- extracting ingredient text using OCR,
- analyzing ingredients using NLP techniques,
- detecting allergens and harmful ingredients,
- and generating personalized risk assessments for users.

Objectives:

The objectives of the project are:

1. To develop a web-based food ingredient analysis system.
2. To extract ingredient text from uploaded food package images using OCR.
3. To preprocess and clean extracted text using NLP techniques.
4. To classify ingredients into intolerance or allergy categories.
5. To provide personalized food safety analysis based on user-selected dietary restrictions.
6. To generate food risk scores and warnings for users.
7. To study and compare machine learning and deep learning models for ingredient classification.

Target Users:

This system is designed to help users with:

User Type	Example
Lactose Intolerance	Milk, whey, casein sensitivity
Gluten/Wheat Allergy	Gluten, wheat, barley sensitivity
Diabetes	Sugar and glucose monitoring
Vegan Users	Avoiding animal-derived ingredients
Nut Allergy	Peanut/tree nut sensitivity
Yeast Intolerance	Fermentation-related ingredients
Salicylate Sensitivity	Salicylate-containing foods
Nickel Sensitivity	Nickel-related food products

Proposed System Overview:

The proposed system workflow is:

Food Package Image
↓
OCR Text Extraction
↓
Text Preprocessing
↓
Ingredient Analysis
↓
Allergen/Restriction Detection
↓
Risk Scoring
↓
User Safety Recommendation

Technologies and Frameworks:

Backend Framework

- Django

Django will manage:

- backend logic,
- authentication,
- database operations,
- OCR integration,
- and ML model execution.

Frontend Technologies

- HTML
- CSS
- JavaScript
- Tailwind CSS

The frontend will provide:

- image upload interface,
- user profile settings,
- food analysis dashboard,
- and result visualization.

OCR and Image Processing:

OCR Libraries

- PaddleOCR
  OR
- EasyOCR

OCR will extract ingredient text from uploaded food package images.

Image Processing

- OpenCV

Used for:

- image enhancement,
- noise removal,
- text sharpening,
- and preprocessing before OCR.

NLP and Machine Learning:

Text Preprocessing Techniques

The system will apply:

- lowercase conversion,
- punctuation removal,
- stop-word removal,
- stemming or lemmatization,
- tokenization,
- and ingredient normalization.

Feature Extraction Techniques:

The following NLP feature extraction methods will be explored:

Technique	Purpose
Bag of Words (BoW)	Word frequency representation
TF-IDF	Important keyword weighting
Word2Vec	Semantic relationship learning

Machine Learning and Deep Learning Models:

The project will evaluate and compare several models:

Model	Purpose
Linear SVM	Ingredient classification
Random Forest	Risk prediction
Dense Neural Network	Deep classification
CNN	Text pattern extraction
LSTM	Sequential ingredient analysis

Additionally, modern transformer-based NLP models may also be explored:

- BERT
- BioBERT

Dataset:

The project may use publicly available food datasets such as:

- Open Food Facts
- USDA FoodData Central

The dataset will contain:

- ingredient labels,
- allergen information,
- food categories,
- and nutritional information.

Custom labels for intolerances and restrictions may also be generated.

Expected features:

The system is expected to provide:

- Food image upload
- OCR ingredient extraction
- Allergy/intolerance detection
- Personalized restriction matching
- Food risk scoring
- Safe/unsafe recommendations
- User profile management
- Ingredient warning visualization

Expected Outcomes:

The final system will:

- assist users in making safer food choices,
- reduce accidental allergen consumption,
- automate ingredient analysis,
- and improve awareness about hidden food ingredients.

The project can also serve as:

- a healthcare assistant tool,

- a dietary monitoring system,
- or a smart food recommendation platform.

What Will Be Learned:

Through this project, the following knowledge and skills will be gained:

Artificial Intelligence & Machine Learning

- NLP techniques
- Text classification
- Feature extraction
- Deep learning models
- OCR integration

Web Development

- Django backend development
- Database management
- Frontend UI development
- API integration

Computer Vision

- Image preprocessing
- OCR systems
- Text extraction from images

Data Science

- Dataset creation
- Data preprocessing
- Model training and evaluation
- Performance analysis

Future Improvements:

Future versions of the system may include:

- barcode scanning,
- mobile application support,

- multilingual ingredient analysis,
- cloud deployment,
- personalized AI recommendations,
- and real-time supermarket product scanning.

Conclusion:

This project proposes an intelligent AI-powered food ingredient analysis system that combines OCR, NLP, and machine learning techniques to help users identify potentially harmful food products according to their allergies, intolerances, and dietary restrictions.

The proposed system aims to improve food safety awareness, simplify ingredient understanding, and provide practical assistance for users in everyday food purchasing decisions.

I believe this project will provide a strong foundational understanding of deep learning applications in food safety analysis, particularly in allergen and intolerance detection using OCR and NLP techniques, and will fulfill the learning objectives of the course.

The proposed system aims to improve food safety awareness, simplify ingredient understanding, and provide practical assistance for users in everyday food purchasing decisions.

Best regards,
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